

**General Characteristics**

141. The decrease in atomic volume from *Cr* to *Cu* is very negligible because  
 (a) Increase in nuclear charge  
 (b) Screening effect  
 (c) Unpaired electrons of *Cr*  
 (d) None
142. The heaviest atom amongst the following is  
 (a) Uranium (b) Radium  
 (c) Lead (d) Mercury
143. Thallium shows different oxidation states because  
 (a) It is a transition metal  
 (b) Of inert-pair effect  
 (c) Of its high reactivity  
 (d) Of its amphoteric character
144. The test of ozone  $O_3$  can be done by  
 (a) *Ag* (b) *Hg*  
 (c) *Au* (d) *Cu*
145. Which of the following set of elements does not belong to transitional elements set  
 (a) *Fe, Co, Ni*  
 (b) *Cu, Ag, Au*  
 (c) *Ti, Zr, Hf*  
 (d) *Ga, In, Tl*
146. The transition metals mostly are  
 (a) Diamagnetic  
 (b) Paramagnetic  
 (c) Neither diamagnetic nor paramagnetic  
 (d) Both diamagnetic and paramagnetic
147. The correct statement in respect of *d*-block elements  
 (a) They are all metals  
 (b) They show variable valency  
 (c) They form coloured ions and complex salts  
 (d) All above statements are correct
148. Which one of the following is an example of non-typical transition elements  
 (a) *Li, K, Na* (b) *Be, Al, Pb*  
 (c) *Zn, Cd, Hg* (d) *Ba, Ca, Sr*
149. Which one is wrong in the following statements  
 (a) Gold is considered to be the king of metals  
 (b) Gold is soluble in mercury  
 (c) Copper is added to gold to make it hard  
 (d) None of these
150. The number of unpaired electrons in  $Cr^{3+}$  ion is  
 (a) 3 (b) 5  
 (c) 4 (d) 1
151. The metal ion which does not form coloured compound is  
 (a) Chromium (b) Manganese  
 (c) Zinc (d) Iron



152. Super alloys are usually  
(a) Iron based  
(b) Nickel based  
(c) Cobalt based  
(d) Based on all of these
153. The transitional metal which shows oxidation state from +2 to +7 belong to group  
(a) VII B (b) VI B  
(c) II B (d) III B
154. Which of the following may be colourless  
(a)  $Cr^{+3}$  (b)  $Cu^{+}$   
(c)  $Fe^{+3}$  (d)  $Cu^{2+}$
155. Which of the following ions is paramagnetic  
(a)  $Cu^{+}$  (b)  $Zn^{+2}$   
(c)  $Ti^{+3}$  (d)  $Ti^{+4}$
156. Which of the following metals absorbs hydrogen  
(a)  $K$  (b)  $Al$   
(c)  $Zn$  (d)  $Pd$
157. Which of the following ions is coloured  
(a)  $Cu^{+}$  (b)  $Cu^{2+}$   
(c)  $Ti^{4+}$  (d)  $V^{5+}$
158. The metal present in  $B_{12}$  is  
(a) Magnesium (b) Iron  
(c) Cobalt (d) Manganese
159. Which metal does not give the following reaction  $M + \text{water or steam} \rightarrow \text{oxide} + H_2$   
(a) Mercury  
(b) Iron  
(c) Sodium  
(d) Magnesium
160. Ionisation potential values of  $d$ -block elements as compared to ionization potential value of  $f$ -block elements are  
(a) Higher (b) Equal  
(c) lower (d) All of these

